Light

GCSE Computer Science — Algorithms: Searching and Sorting

Over **1,100,000+ marks** of exam-aligned content across **67 qualifications**. 4,200+ auto-marked assessments & long-answer practice. No login required.

GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 51 marks | Answered: 0 / 30

00:00 Start Pause Reset ← Back to Algorithms Searching Sorting

Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

An algorithm checks each item in a list one by one, starting from the first item, until it finds the item being searched for.
What is this algorithm called?

- ○ **A.** Binary search
- ○ **B.** Linear search
- ○ **C.** Bubble sort
- ○ **D.** Merge sort

Q2 (1 mark)

An algorithm repeatedly divides a sorted list in half, comparing the middle element with the target value to determine which half to search next.

What is this algorithm called?

- ○ **A.** Binary search
- ○ **B.** Linear search
- ○ **C.** Bubble sort
- ○ **D.** Merge sort

Q3 (1 mark)

A sorting algorithm repeatedly compares adjacent pairs of items in a list, swapping them if they are in the wrong order, until no more swaps are needed.
What is this algorithm called?

- ○ **A.** Linear search
- ○ **B.** Binary search
- ○ **C.** Bubble sort
- ○ **D.** Merge sort

Q4 (1 mark)

A sorting algorithm divides a list into smaller sublists until each contains one item, then combines them back together in sorted order.
What is this algorithm called?

- ○ **A.** Linear search
- ○ **B.** Bubble sort
- ○ **C.** Binary search

- ○ **D.** Merge sort

Q5 (1 mark)

Which condition must be true before binary search can be used on a list?

- ○ **A.** The list must contain only numbers
- ○ **B.** The list must have an even number of items
- ○ **C.** The list must contain no duplicate values
- ○ **D.** The list must be sorted

Q6 (1 mark)

A list of names is stored in no particular order. Which searching algorithm can be used to find a specific name?

- ○ **A.** Only binary search
- ○ **B.** Only linear search
- ○ **C.** Either linear search or binary search
- ○ **D.** Neither algorithm can search unsorted data

Q7 (1 mark)

In bubble sort, after the first complete pass through a list of 5 items, which statement is true?

- ○ **A.** The smallest item is in its correct position

- ○ **B.** All items are in their correct positions
- ○ **C.** No items have moved
- ○ **D.** The largest item is in its correct position

Q8 (1 mark)

In merge sort, how many items does each sublist contain after the list has been fully divided?

- ○ **A.** Two items each
- ○ **B.** Half the original number of items
- ○ **C.** One item each
- ○ **D.** Zero items each

Q9 (1 mark)

A linear search is performed on a list of 100 items. In the worst case, how many comparisons are needed?

- ○ **A.** 1
- ○ **B.** 50
- ○ **C.** 100
- ○ **D.** 200

Q10 (1 mark)

A sorted list contains 8 items. After the first comparison in binary search, how many items remain in the search area?

- ○ **A.** 8
- ○ **B.** 7
- ○ **C.** 4
- ○ **D.** 1

Q11 (1 mark)

Which searching algorithm is generally faster when searching a large sorted list?

- ○ **A.** Linear search, because it starts from the beginning
- ○ **B.** Both are equally fast on sorted data
- ○ **C.** Neither algorithm works on sorted data
- ○ **D.** Binary search, because it eliminates half the data with each comparison

Q12 (1 mark)

How does bubble sort know that the list is fully sorted and can stop?

- ○ **A.** When a complete pass occurs with no swaps
- ○ **B.** When the first item is the smallest
- ○ **C.** When all items have been compared once

- ○ **D.** When the list has been reversed

Q13 (1 mark)

A binary search is performed on the sorted list [5, 12, 18, 25, 33, 41, 50] to find the value 33.
Which values are compared with 33 during the search?

- ○ **A.** 5, 12, 18, 25, 33
- ○ **B.** 25, 41, 33
- ○ **C.** 25, 33
- ○ **D.** 50, 25, 33

Q14 (1 mark)

A list of 6 items is being sorted using bubble sort.
What is the maximum number of passes needed to fully sort the list?

- ○ **A.** 6
- ○ **B.** 5
- ○ **C.** 12
- ○ **D.** 36

Q15 (2 marks)

The list [8, 3, 5, 1] is sorted using merge sort.
What are the sublists after the first division?

- A. [8] [3] [5] [1]
- B. [8, 3, 5] [1]
- C. [1, 3, 5, 8]
- D. [8, 3] and [5, 1]

Q16 (2 marks)

A sorted list contains 1000 items. Using binary search, what is the maximum number of comparisons needed to find any item or confirm it is not present?

- A. 100
- B. 10
- C. 500
- D. 1000

Q17 (2 marks)

A list contains 5 items. How many comparisons are made during the first pass of bubble sort?

- A. 4
- B. 5
- C. 10
- D. 25

Q18 (2 marks)

A school stores student records in the order they enrolled, with new students added at the end. The school needs to search for students by name.
Which searching algorithm is most appropriate?

- ○ **A.** Binary search, because it is faster
- ○ **B.** Linear search, because the data is unsorted by name
- ○ **C.** Merge sort, then binary search
- ○ **D.** No algorithm can search this data

Q19 (2 marks)

Two sorted sublists [1, 4, 7] and [2, 5, 6] are being merged in merge sort. What is the order of the merged list after all items are combined?

- ○ **A.** [1, 2, 4, 5, 6, 7]
- ○ **B.** [1, 4, 7, 2, 5, 6]
- ○ **C.** [2, 5, 6, 1, 4, 7]
- ○ **D.** [7, 6, 5, 4, 2, 1]

Q20 (2 marks)

In which situation would linear search find an item using only one comparison?

- ○ **A.** When the item is in the middle of the list
- ○ **B.** When the item is at the end of the list

- ○ **C.** When the item is at the start of the list
- ○ **D.** When the list contains only duplicate values

Q21 (2 marks)

In an optimised bubble sort, each pass examines fewer items than the previous pass.
Why is this optimisation valid?

- ○ **A.** Items at the start are always smaller
- ○ **B.** Items at the end are already in their correct positions after each pass
- ○ **C.** The algorithm becomes faster by skipping random items
- ○ **D.** Comparisons are not needed after the first pass

Q22 (2 marks)

Merge sort is often preferred over bubble sort for large lists.
Why is merge sort generally more efficient?

- ○ **A.** Merge sort uses less memory
- ○ **B.** Merge sort makes fewer comparisons overall
- ○ **C.** Merge sort does not require any comparisons
- ○ **D.** Merge sort only works on small lists

Q23 (2 marks)

A binary search is performed on the sorted list [10, 20, 30, 40, 50] to find the value 25.
What happens during the search?

- ○ **A.** The value 25 is found at position 2
- ○ **B.** The search compares 30, then 10, then 20, and concludes 25 is not in the list
- ○ **C.** The search compares 30, then 20, and concludes 25 is not in the list
- ○ **D.** Linear search is used instead because 25 is not present

Q24 (2 marks)

The list [6, 2, 8, 4] is sorted using bubble sort.
What does the list look like after exactly two complete passes?

- ○ **A.** [2, 4, 6, 8]
- ○ **B.** [2, 6, 4, 8]
- ○ **C.** [2, 4, 8, 6]
- ○ **D.** [6, 2, 4, 8]

Q25 (2 marks)

The list [5, 2, 8, 1] is sorted using merge sort.
Which sequence correctly shows the divide and merge phases?

- ○ **A.** Divide: [5,2,8,1] → [5,2] [8,1] → [5] [2] [8] [1]. Merge: [2,5] [1,8] → [1,2,5,8]
- ○ **B.** Divide: [5,2,8,1] → [5] [2] [8] [1]. Merge: [1,2,5,8]

- ○ **C.** Divide: [5,2,8,1] → [5,2,8] [1]. Merge: [1,2,5,8]
- ○ **D.** Divide: [5,2,8,1] → [2,5] [1,8]. Merge: [1,2,5,8]

Q26 (3 marks)

A binary search is performed on the sorted list [3, 7, 11, 15, 19, 23, 27, 31] to find the value 19.
Which sequence of comparisons occurs?

- ○ **A.** 15, 23, 19
- ○ **B.** 15, 19
- ○ **C.** 3, 7, 11, 15, 19
- ○ **D.** 31, 23, 19

Q27 (3 marks)

A library system stores 50,000 books sorted alphabetically by title. Staff frequently search for specific titles.
Which search algorithm should be used and why?

- ○ **A.** Linear search, because it is simpler to implement
- ○ **B.** Bubble sort followed by linear search
- ○ **C.** Linear search, because binary search cannot handle text data
- ○ **D.** Binary search, because the data is sorted and binary search needs at most about 16 comparisons instead of up to 50,000

Q28 (3 marks)

A teacher is explaining when to use bubble sort versus merge sort. Which statement correctly compares the two algorithms?

- ○ **A.** Bubble sort is always faster than merge sort regardless of list size
- ○ **B.** Merge sort is always faster than bubble sort regardless of list size
- ○ **C.** Both algorithms perform identically on all list sizes
- ○ **D.** Bubble sort may be acceptable for small lists and is simpler to understand; merge sort is significantly faster for large lists but uses additional memory and is more complex

Q29 (3 marks)

The list [5, 3, 4, 2, 1] is sorted using bubble sort.
How many swaps occur during the first complete pass?

- ○ **A.** 2
- ○ **B.** 3
- ○ **C.** 4
- ○ **D.** 5

Q30 (3 marks)

A binary search is performed on the sorted list [2, 4, 6, 8, 10, 12] to find the value 2.
Which sequence of comparisons occurs before the value is found?

- ○ **A.** 6, 2
- ○ **B.** 6, 4, 2

- **C.** 8, 4, 2
- **D.** 12, 6, 2

Total: **51** marks • Time: **30 min**

↑ Back to top

Submit answers

Williams Physics Education — over **1,100,000+ marks** of exam-aligned content and **4,200+ auto-marked assessments** across 67 qualifications at GCSE, A-Level, AP & Professional Qualifications. Free to use, no login required.

About & Contact • Privacy • Terms • Accessibility • Copyright • DMCA • YouTube

WPE is free for everyone. If our resources help you or your students, you can support the project here.

© 2026 Williams Physics Education. All rights reserved.